25901093

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU–UGP (HONOURS) Regular Examination – March 2026
Fourth Semester
Skill Enhancement Course (SEC)

MG4SECBCH200 – Biochemical Tests in Disease Diagnosis

(2024 Admission Onwards)

Duration: 1 Hour 30 Minutes **Maximum Marks:** 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer all questions
Each question carries **1 mark**

1. Which ethical responsibilities uses external quality assessment.
   ANS: a) Quality Assurance and Ethical Practice.**[U] / [CO1]**
2. The normal 24-hour urinary uric acid excretion is
   ANS: c) 250-750 mg/dl **[An] / [CO3]**
3. Which test reflects long-term blood glucose control in diabetes?
   ANS: b) HbA1c **[K] / [CO4]**
4. Which of the following is not a component of a routine blood test?
   ANS: c) Urea **[U] / [CO1]**
5. Elevated bilirubin levels generally indicate:
   ANS: c) Liver dysfunction **[U] / [CO3]**

[1×5 = 5 marks]

Part B

Fill in the Blank Questions
Answer all questions
Each question carries **2 marks**

6. Blood specimens must be transported in _____containers to prevent leakage.
   ANS:leak-proof **[U] / [CO1]**

7. Type 2 diabetes mellitus is mainly due to _____ resistance.
   ANS: Insulin **[K] / [CO2]**

8. A device that uses a biological component linkage to a transducer to convert biochemical reactions into measurable signals is _____.

ANS: Biosensors **[U] / [CO4]**

9. The first morning urine sample is preferred for protein and chorionic gonadotropin measurement because it is more _____.
   ANS: Concentrated **[U] / [CO1]**
10. The normal albumin –globulin (A/G) ratio ranges from _____ to _____ng/ml.
   ANS: 0.8 – 2 ng/ml **[E] / [CO3]**

[2×5 = 10 marks]

Part C

Short Answer Questions

Answer any 5 questions

Each question carries **3 marks**

11. Define analytical factors in clinical laboratory testing.

ANS: Define Analytical factors**.** **(1 mark)**

Laboratory preparation of Samples, Standards, procedural controls, I nstrumentperformance, reagent quality, calibration, analytical methods, quality control, environmental conditions, and personnel competency.**(any 4 points - 2 Marks)**

[K] / [CO1]

12 Label the main advantage of HbA1c testing.

ANS:

- Reflects long-term glycemic control (past 2–3 months).
- No fasting required; can be done anytime.
- Not affected by short-term glucose fluctuations.
- Good predictor of diabetic complications.
- Standardized and reliable for diagnosis and monitoring. **(any 3 points)**

[K] / [CO2]

13 A patient presents with itching and wheezing. DC shows eosinophils = 9%. Analyse the condition.

ANS: Symptoms indicates Allergic reactions – 1 marks

Normal range of eosinophils (1- 6%) – Interpretation – 2 marks

[An] / [CO4]

14 Describe the normal volume and colour of CSF and explain their Clinical relevance.

ANS: Normal volume of CSF – 125 ml. (1 Mark)

Normal colour of CSF - Clear and colorless. (1 Mark)

Helps in diagnosis of infections, Bleeding, Neurological Disorders and Malignancies. (any 2 points – 1 mark)

[U] / [CO1]

15 Analyse the changes in AFP levels from fetal life to adulthood and their diagnostic importance.

ANS:Fetallife : AFP level is very high (1 mark)

After birth : AFP decreases to normal (1 Mark)

Diagnostic importance - Cancer diagnosis, Pregnancy screening (1 mark)

[An] / [CO3]

16 Analyse the reason why serum creatinine is considered a reliable indicator of GFR

ANS:

- Creatinine is produced at a constant rate from muscle metabolism (breakdown of creatine phosphate).
- This steady production makes it a stable marker under normal physiological conditions.
- Creatinine is freely filtered through the glomerular membrane without restriction.
- The concentration serum creatinine increases with decrease in GFR.
- Even a small decrease in GFR causes a significant rise in serum creatinine.

(any 3 point -3 marks) **[An] / [CO3]**

17 Explain internal quality control.

ANS:

Explain quality control. – 1 mark

Write a note on internal quality control - 2 marks

[U] / [CO1]

[3×5 = 15 marks]

Part D – Short Essay Questions

Answer any 4 questions

Each question carries **5 marks**

18. Evaluate the clinical significance of Triglyceride.

ANS: What is Triglyceride – 1 mark.

Normal range of TG – 1 mark.

Clinical Significance of Triglyceride – 3 marks. **[E] / [CO1]**

19. Describe the Cyanmethemoglobin (Drabkin's) method for estimation of hemoglobin.

ANS: What is haemoglobin -1 mark.

Explain Principle and Procedure of Cyanmethemoglobin method -4 marks.

[U] / [CO1]

20. Describe the role of Glucose Challenge Test in screening for gestational diabetes mellitus.

ANS: What is GCT-1 marks

Explan the importance of GCT in pregnancy– 4 marks

[U] / [CO2]

21. Summarize the physiological principles underlying renal function tests and state their clinical applications.

ANS: What is RFT – 1marks

Explain the clinical applications of RFT- 4 marks

[U] / [CO3]

22. Explain how robotic pipetting improve laboratory outcomes.

ANS : Explain Robotics in biochemical testing : 1 marks

- Robotic pipetting significantly enhances laboratory outcomes by improving accuracy, precision, speed, and high throughput.
- Reducing human error and operational costs. It is especially valuable in modern high-volume and high-accuracy diagnostic laboratories. – 4 marks.

[U] / [CO4]

23. Outline next-generation sequencing (NGS).

ANS: What isNext Generation sequencing – 1 mark
Importance of NGS – 1.5 marks
Application of NGS -2.5 marks

[An] / [CO4]

[5×4 = 20 marks]